

ME 350 Lecture #2

Introduction to Linkages

Mechanism Definitions,
Transmission Angle, Mechanical Gain
Kinematic Diagrams, Mobility,
Types of Linkages,
Grashof and Non-Grashof

Prof. Diann Brei

Mechanism Madness!

In a group of 2 to 3 people, make a quick “rough” four bar mechanism

You have 5 mins – so be quick

Helpful steps:

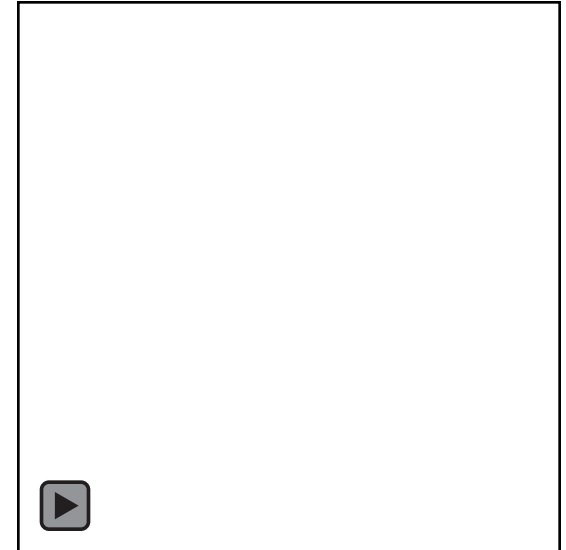
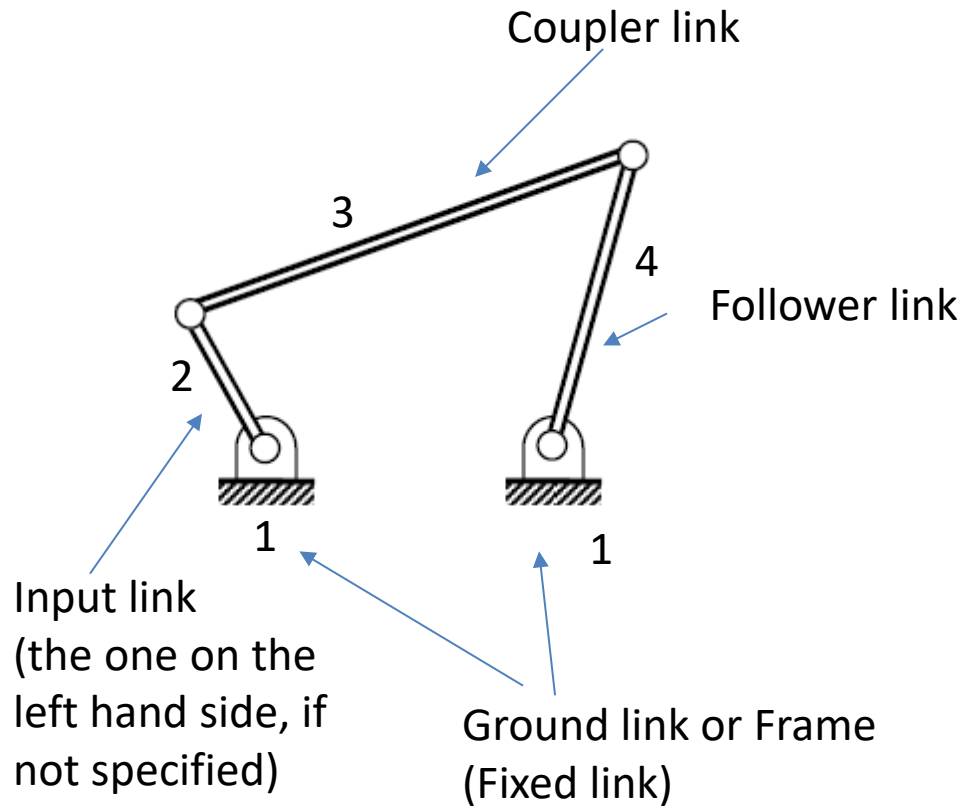
- 1. Gather:** (please only take the following):
 - One yellow, one pink and one green piece of paper
 - 4 brad pins
- 2. Make your “linkages”:** Rip from the paper one “linkage” from each color paper. You should have three different color paper “linkages”. These can be any shape, any size.
Tip – go bigger than smaller.
- 3. Assemble the 4 bar linkage:** Using the 4 brads as connectors - assemble your linkage
 - Connect the one end of the pink link to one end of the yellow link.
 - Connect the other end of the yellow link to one end of the green link
 - Use the table as your last link: Put a brad at the end of the pink link and one at the end of the green link. Tape gently (so you can get off later) only the leg of the brad to the table so that the links can still rotate.

Brads 101

Use the point to punch gently through the paper – don’t get too close to the end of an edge. Rotate the brad a little to make the hole in the paper. Bend the ends but leave room between the papers so that the paper can move relative to each other

Carefully
Make sure that your mechanism moves in some way without ripping

Terminology of a Four-Bar Linkage



Always label ground link as “1”

Linkage

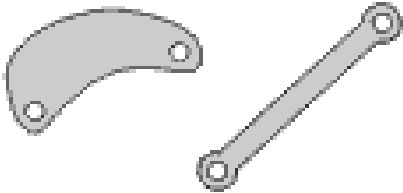
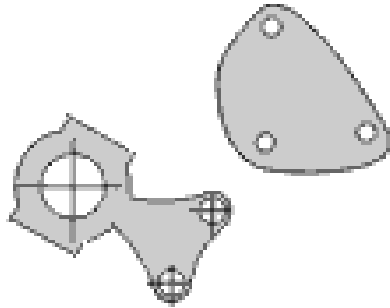
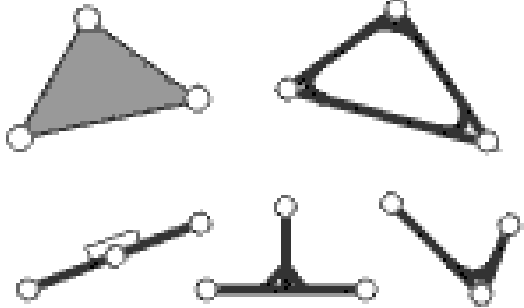
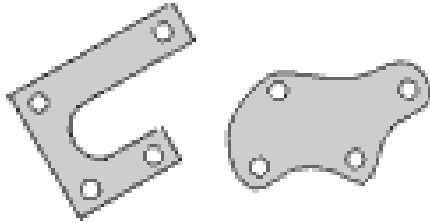
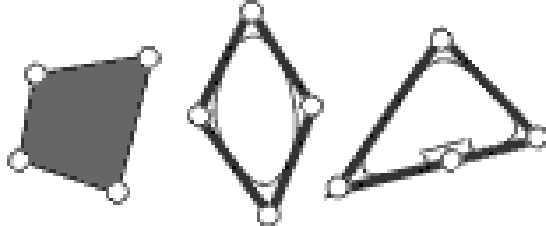
Linkage: an assembly of **links** connected by **joints**

Link: a rigid body with at least two joints

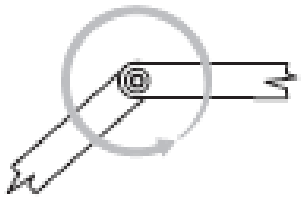
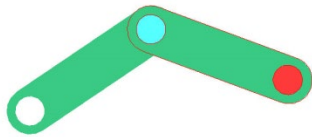
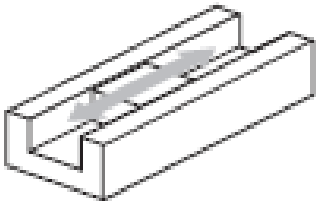
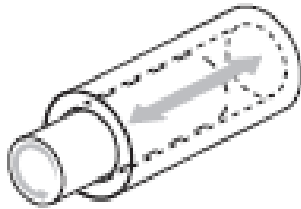
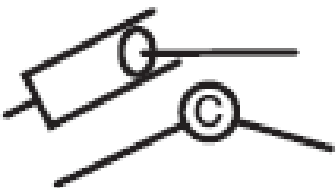
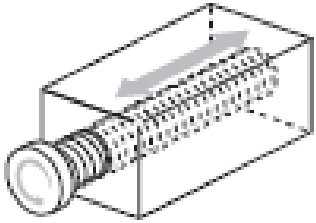
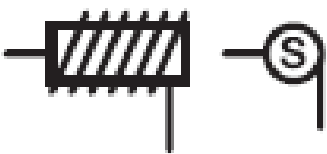
Joint: the connection between two links



Link: a rigid body with at least two joints (points of attachment to other links)

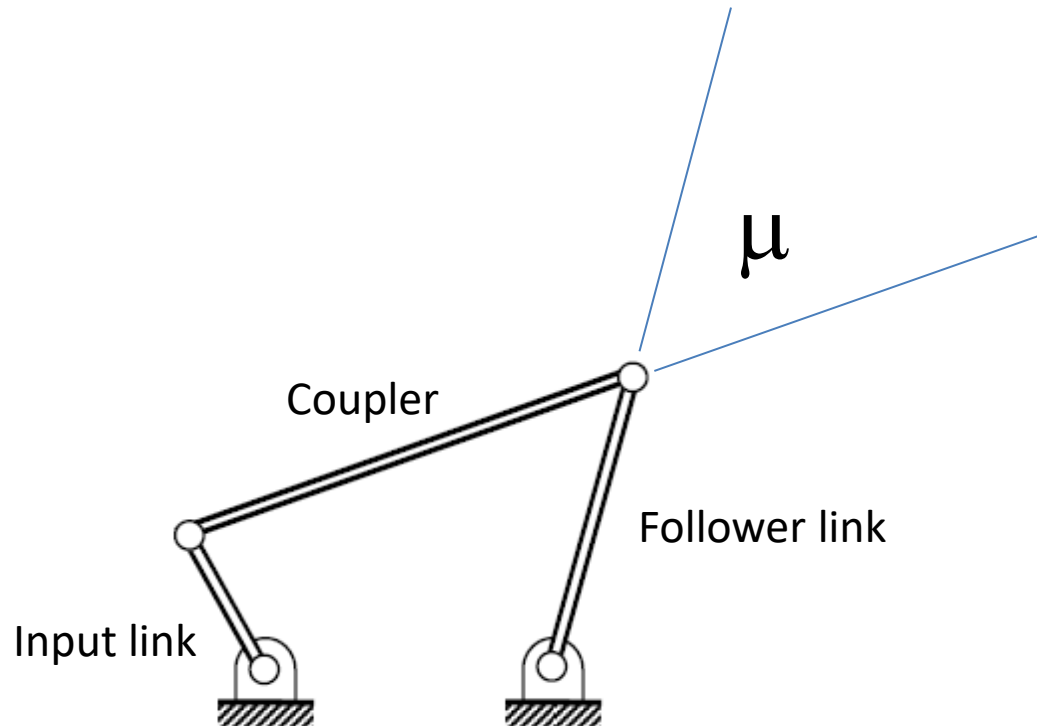
Link Types	Physical forms	Kinematic Representations
Binary		
Ternary		
Quaternary		

Some basic Types of Joints

Joint Type	Depiction	Schematic	Example	Degrees of Freedom
Pin Revolute				1 Rotational
Slider Prismatic				1 Translational
Slider Cylindrical				2 1-Rotational 1-Translational
Screw				1 1-Rotational or 1-Translational (not independent motions)

Transmission Angle, μ

The transmission angle is a means of determining the effectiveness of transferring motion and force to the output link.



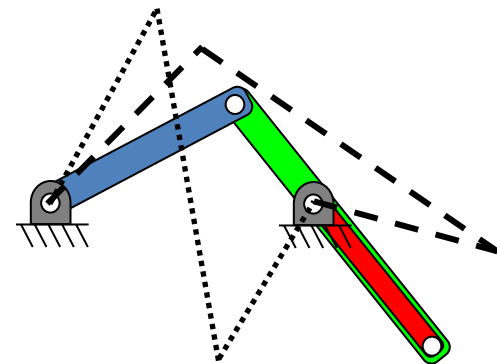
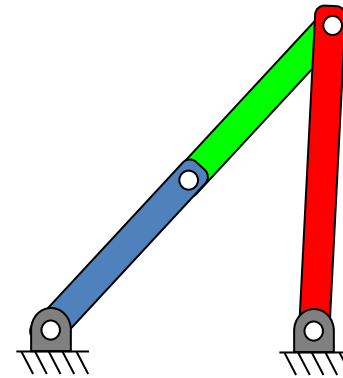
It is defined as the acute angle between the coupler and the follower.

Poor Transmission Angle

- Ideal range of transmission angle
$$\mu \geq 30^\circ$$
$$(\mu \leq 150^\circ)$$
- The following may occur if transmission angle is poor:
 - Mechanism binds (no motion results)
 - Joints fail (pieces go flying)
 - Wrong motion (can't control which way it comes out of the toggle position)

Toggle positions

- Input link collinear with coupler
 - Infinite Mechanical Advantage (theoretically)
 - μ not necessarily = 0
- Coupler collinear with output link (follower)
 - “Change point” (dead center)
 - $\mu = 0$
 - Can’t control which way it comes out of the toggle position



Application Categories

- **Function Generation:**

The correlation of an input function with an output function in a mechanism. Relative motion between links is of interest. It does not require path tracer point on coupler.

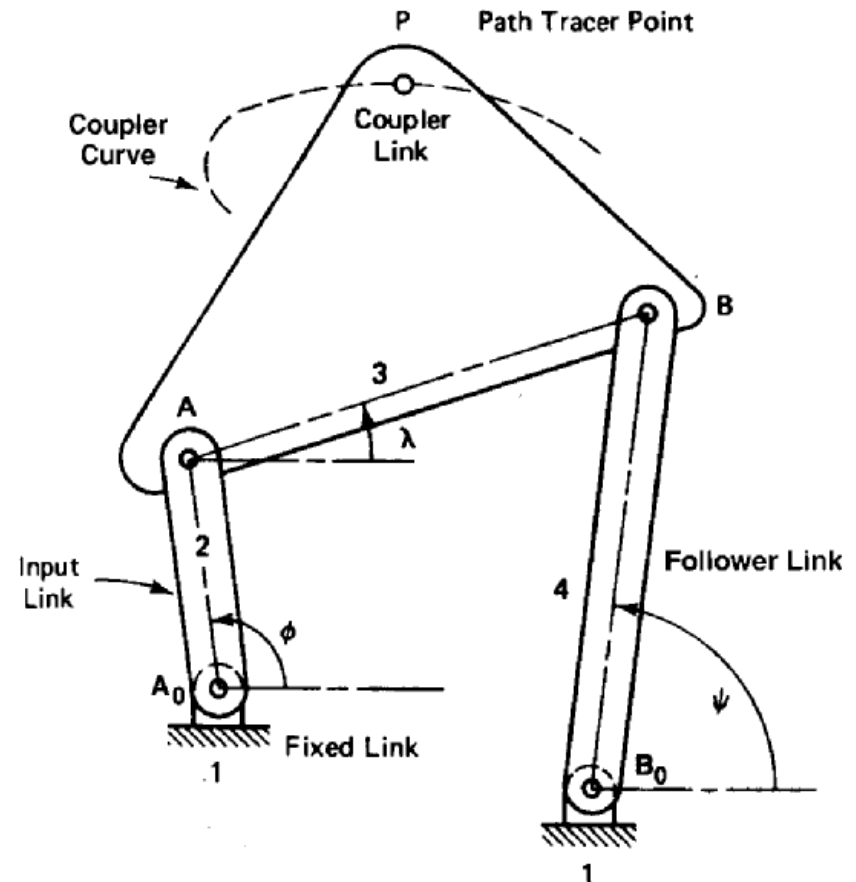
- **Path Generation:**

The control of a point in the plane such that it follows a prescribed path. Only the path of trace point is of interest, not the rotation (orientation) of the coupler link

- **Motion Generation:**

The control of a line in the plane such that it assumes some sequential set of prescribed positions. Entire motion of coupler link is of interest – path tracer point coordinates and angular orientation of couple link.

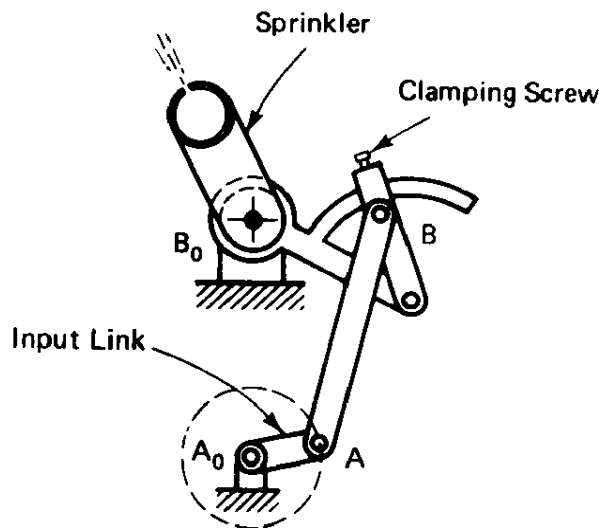
** Note that the points, or positions, prescribed for successive locations of the output (coupler or rocker) link in the plane are generally referred to as *precision points* or *precision positions*.



Function Generator Example:

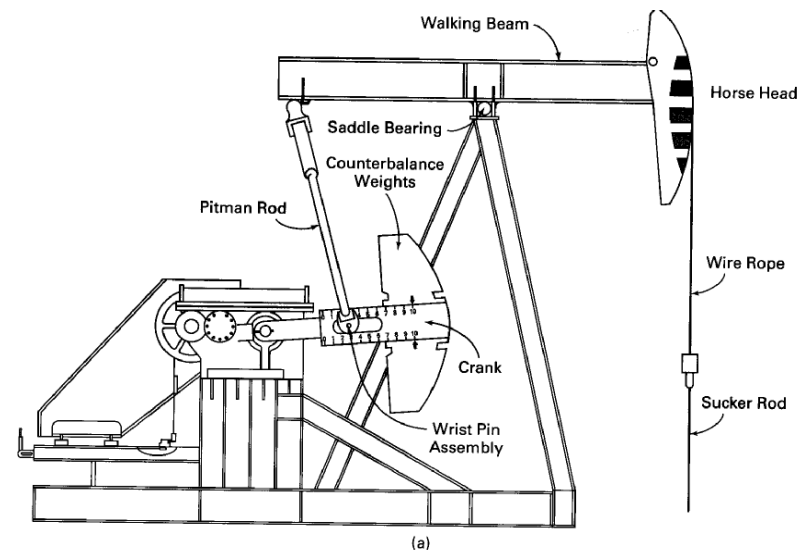
Sprinkler:

Generates function for sprinkling and the range of oscillation of sprinkler head is variable



Oil Rig:

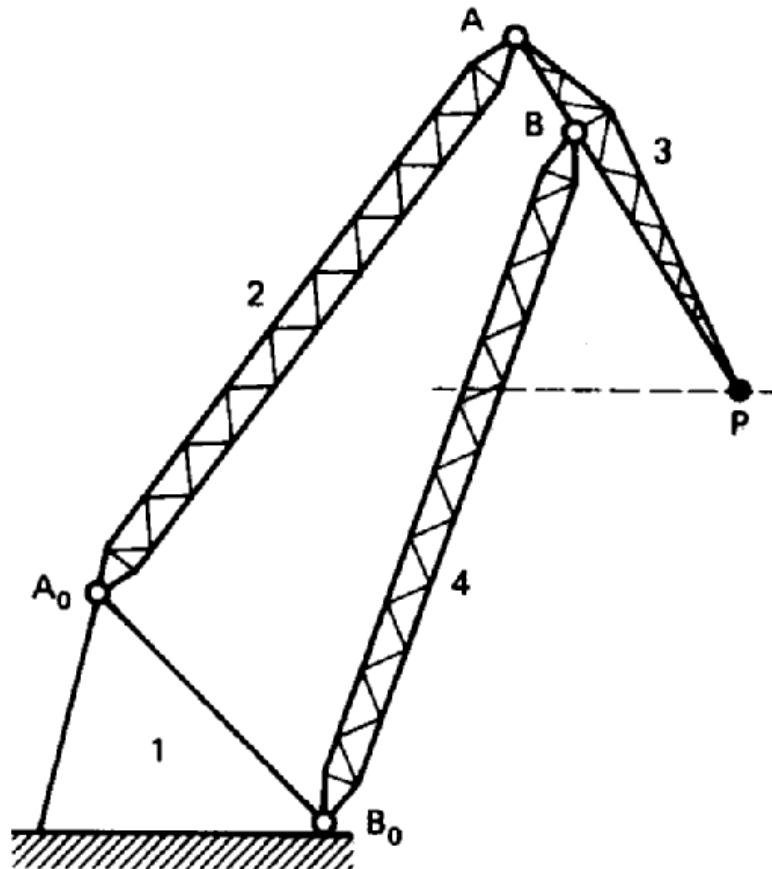
360° rotation of crank is converted into prescribed oscillation of walking beam



Path Generator Example

Crane:

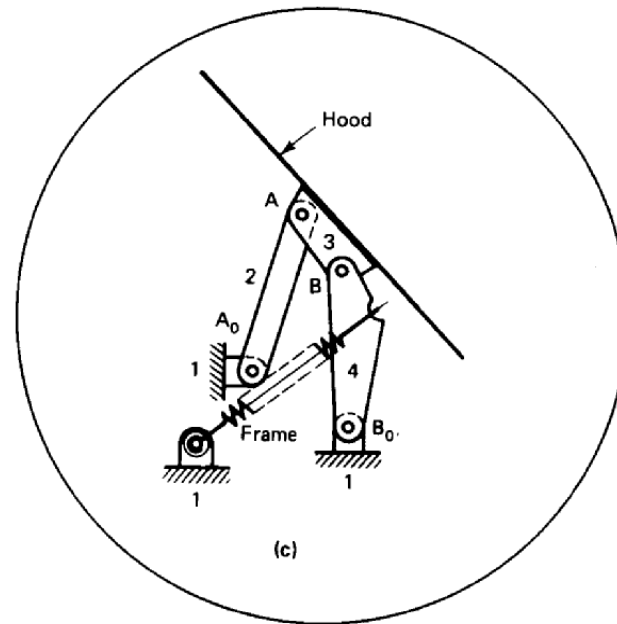
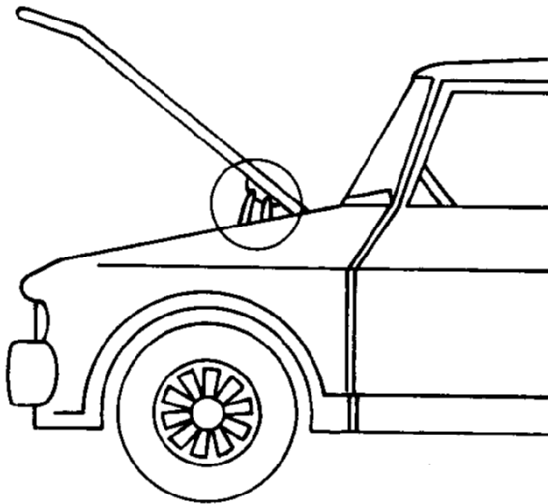
Generates approximate straight-line motion at path tracer point P.



Motion Generation Example:

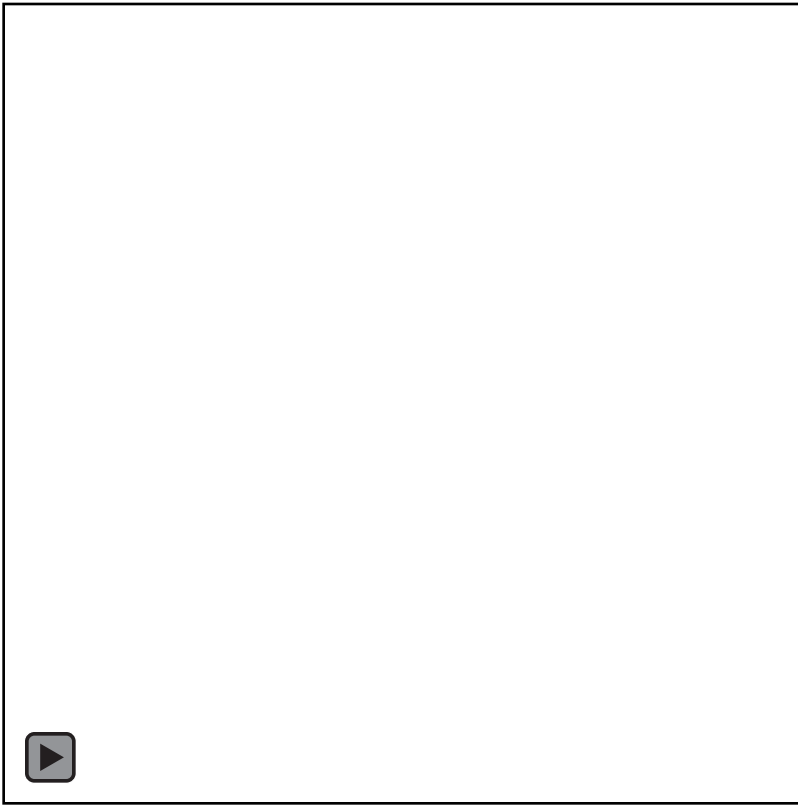
Hood:

Linkage controls relative orientation between hood and car frame. Coordinates of path tracer point on end of hood and angle of hood with car are important.



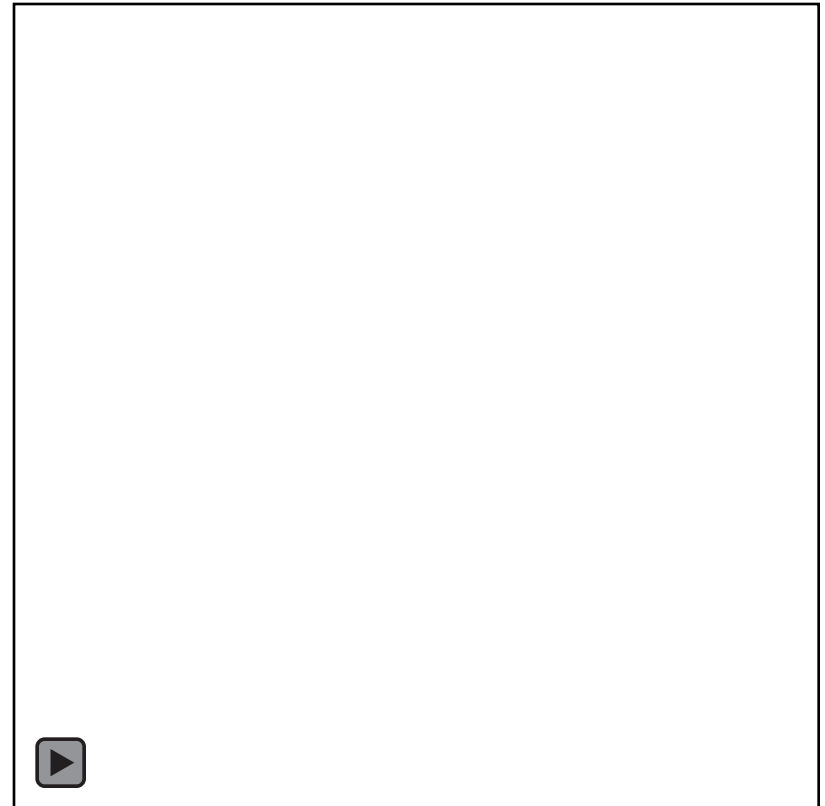
Planar vs. Spatial Mechanisms

Planar (2-D): Links move in parallel planes throughout the motion cycle.



Planar mechanism

Spatial (3-D): At least one link does not move parallel to a single plane.



Spatial mechanism

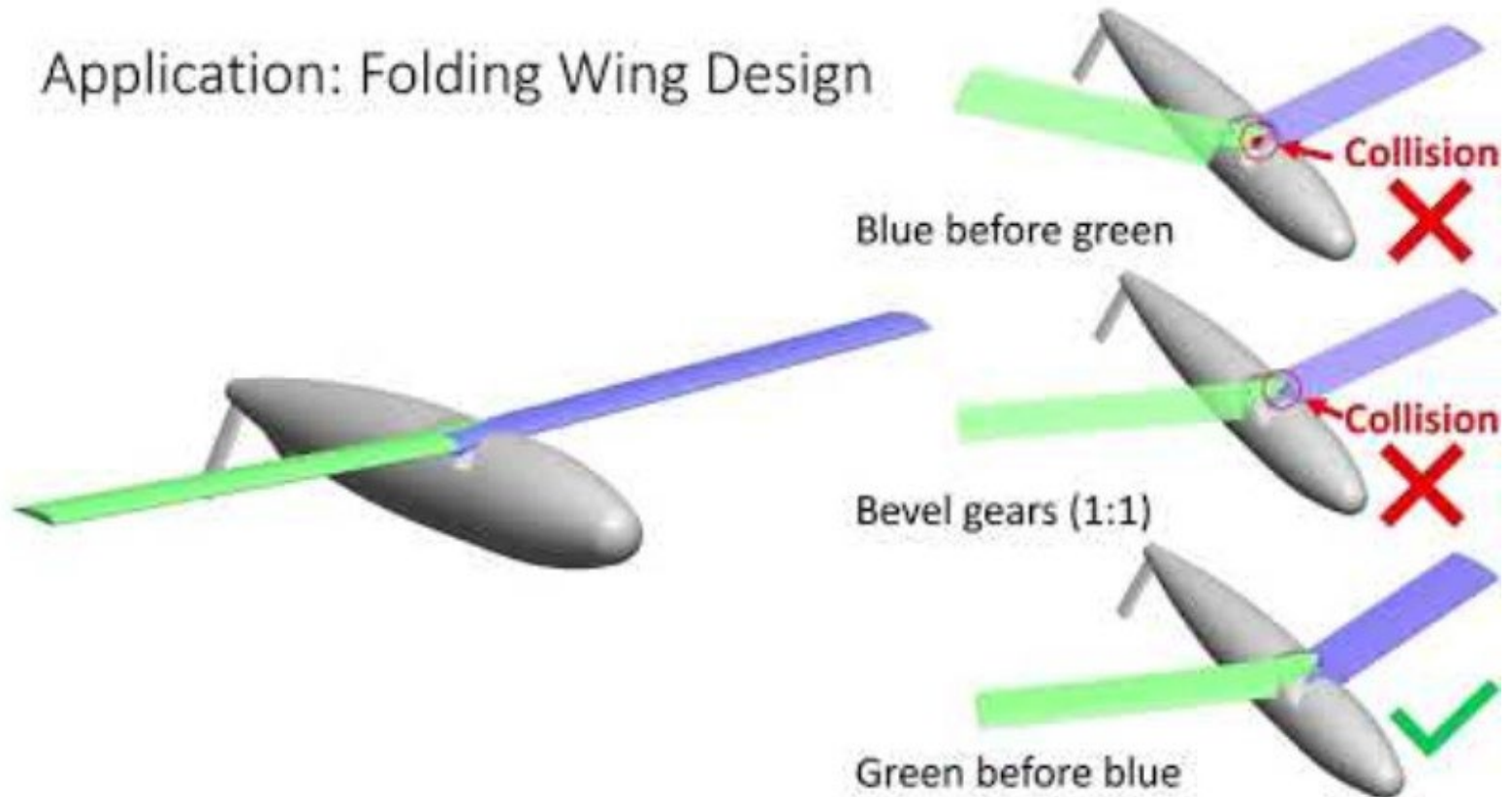
Convertible Sofa-Bed



Planar Mechanism



Application: Folding Wing Design



<https://www.youtube.com/watch?v=bAGA6bvyevU&t=5s>

Folding Stroller



<http://www.youtube.com/watch?v=vG5-mlt58jM>



Take a
quick
minute

Gruebler-Kutzbach Criterion for calculating Mobility (DOF) of a Planar Mechanism

$$M = 3(L - 1) - 2 J_1 - J_2$$

- M (mobility) is the number of independent inputs required to completely specify the geometric configuration of a mechanism
- L is the number of links
- J_1 is the number of joints with one DOF
- J_2 is the number of joints with two DOF

Same equation, using the terminology
and variables from the
Sandor/Erdman/Kota textbook

$$F = 3(n-1) - 2 f_1 - f_2$$

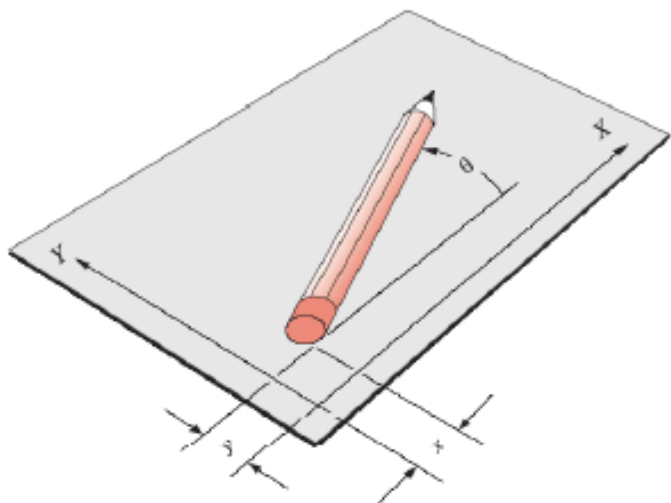
F is the degrees of freedom of the mechanism (mobility)

n is the number of links

f_1 is the number of joints that allow one degree of freedom

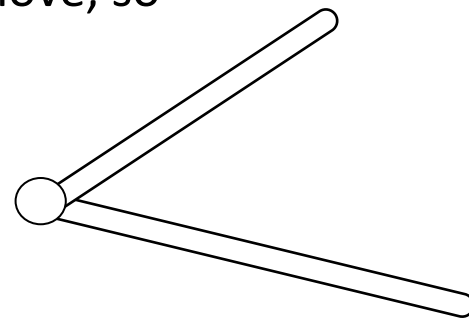
f_2 is the number of joints that allow two degrees of freedom

Derivation of Gruebler-Kutzbach Criterion



Each link has 3 DOF (two translations and one rotation), when moving in a plane.

The ground link cannot move, so subtract one link.



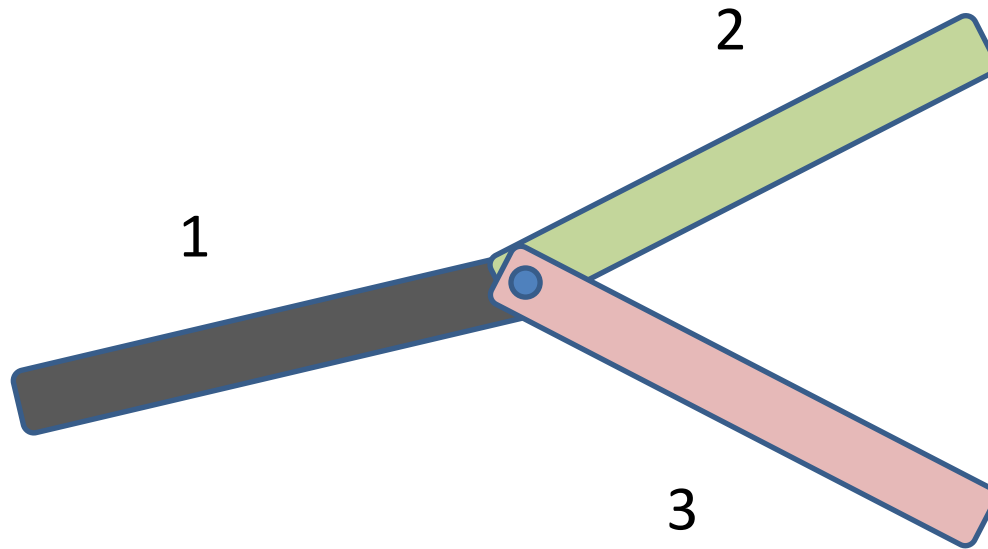
Each J_1 joint allows 1 DOF (removes 2), from one of the links.

Each J_2 joint allows 2 DOF (removes 1), from one of the links.

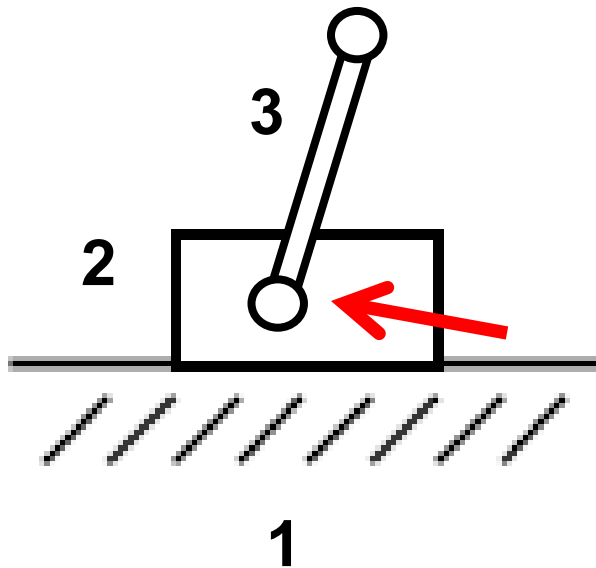
$$M = 3(L - 1) - 2J_1 - J_2$$

Beware!

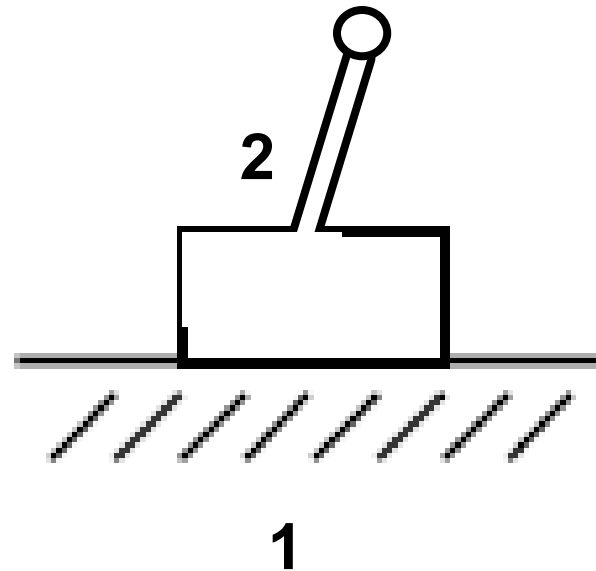
When three links are connected to one pin, you need to count that joint as two separate J1 joints. Otherwise, your calculation of DOF (using the Gruebler – Kutzbach Criterion) will be incorrect.



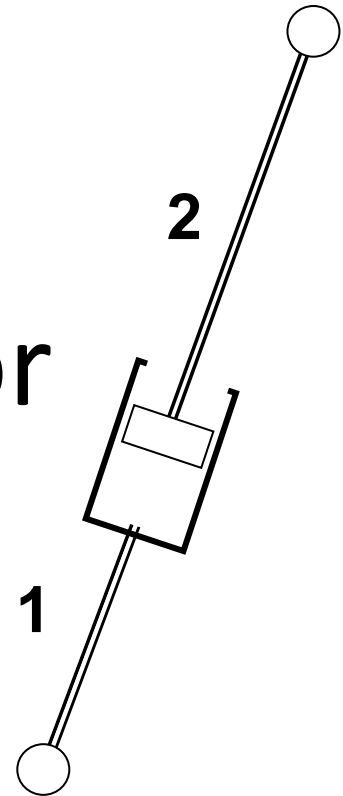
If a slider has a pin joint, then the slider counts as a separate link.



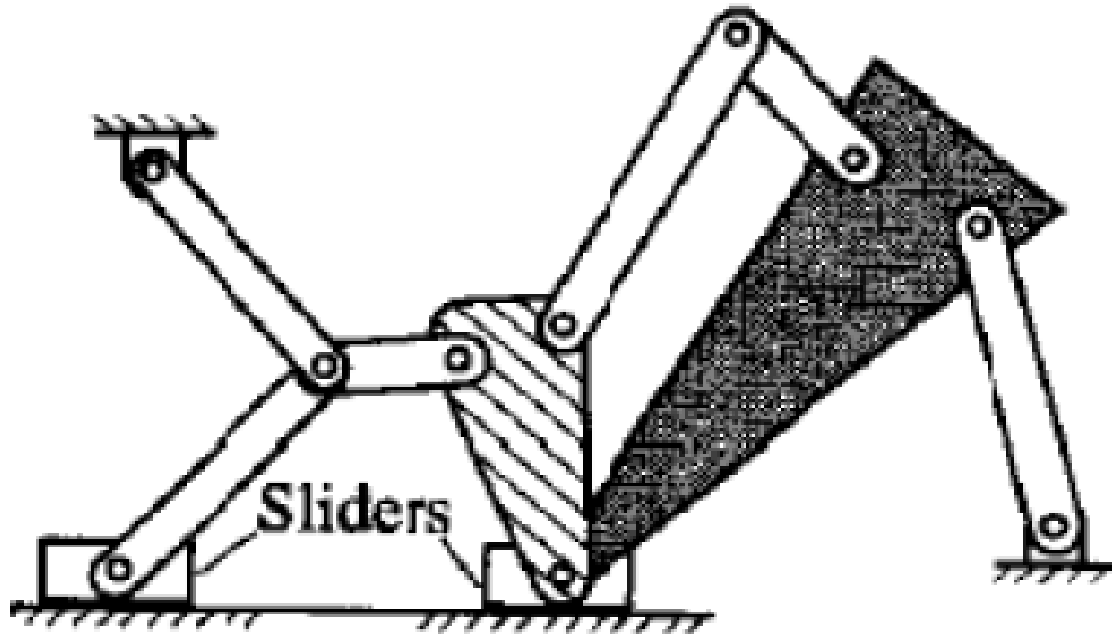
VS.



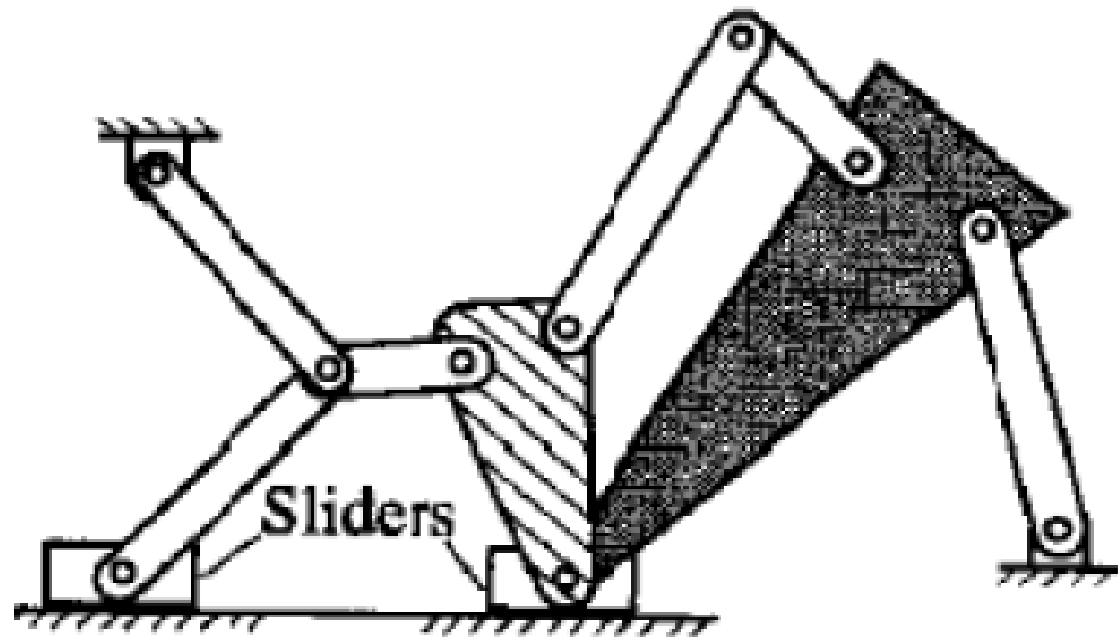
or



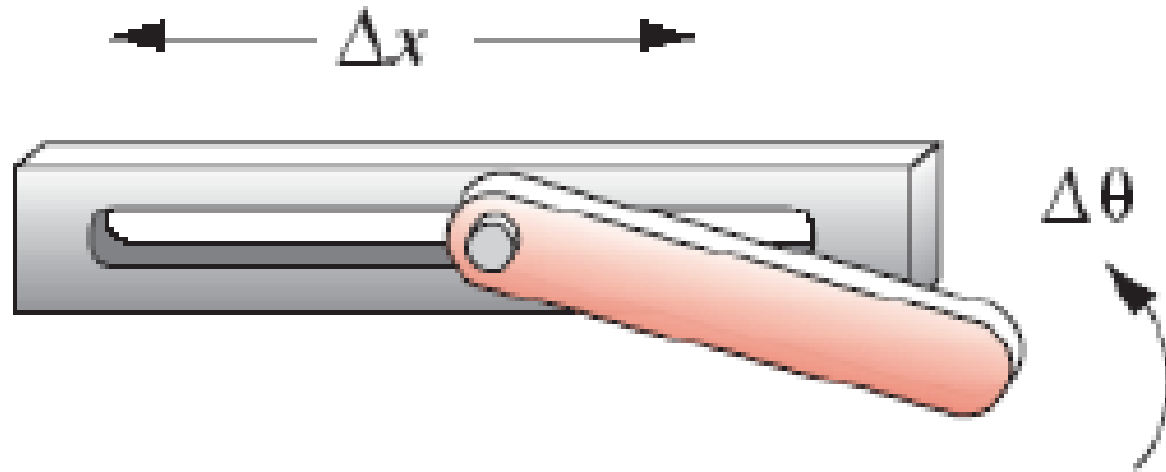
$$M = 3(L - 1) - 2J_1 - J_2$$



- Using the Gruebler-Kutzbach criterion, calculate the mobility of this mechanism.
- Does the Gruebler-Kutzbach criterion tell us where to put the motors?

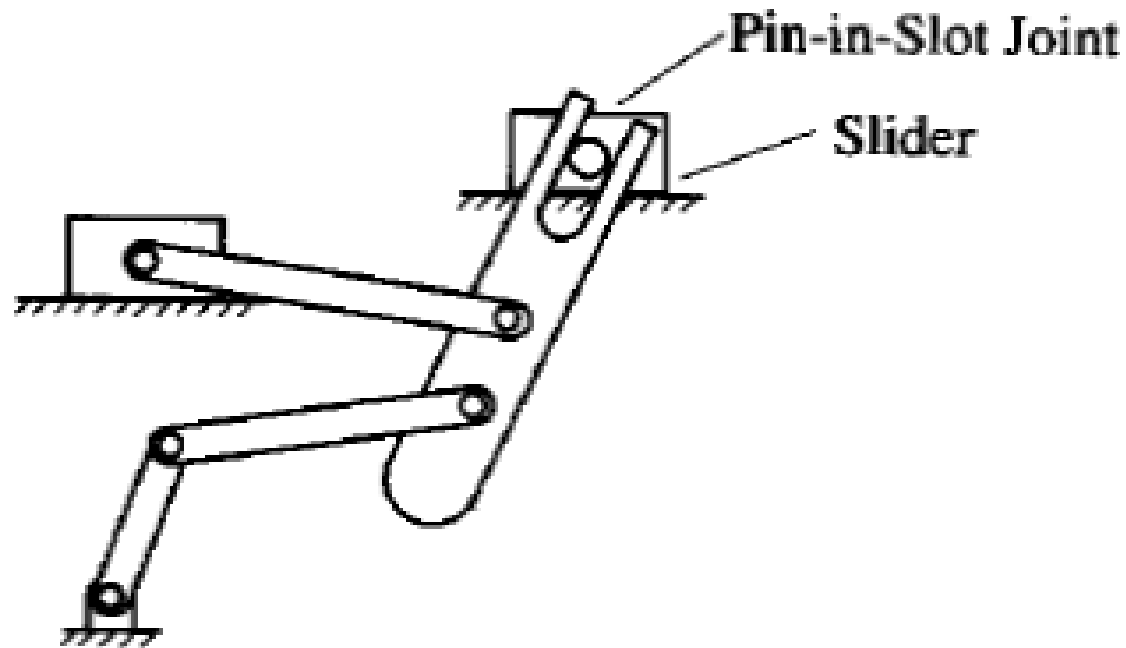


Pin in Slot



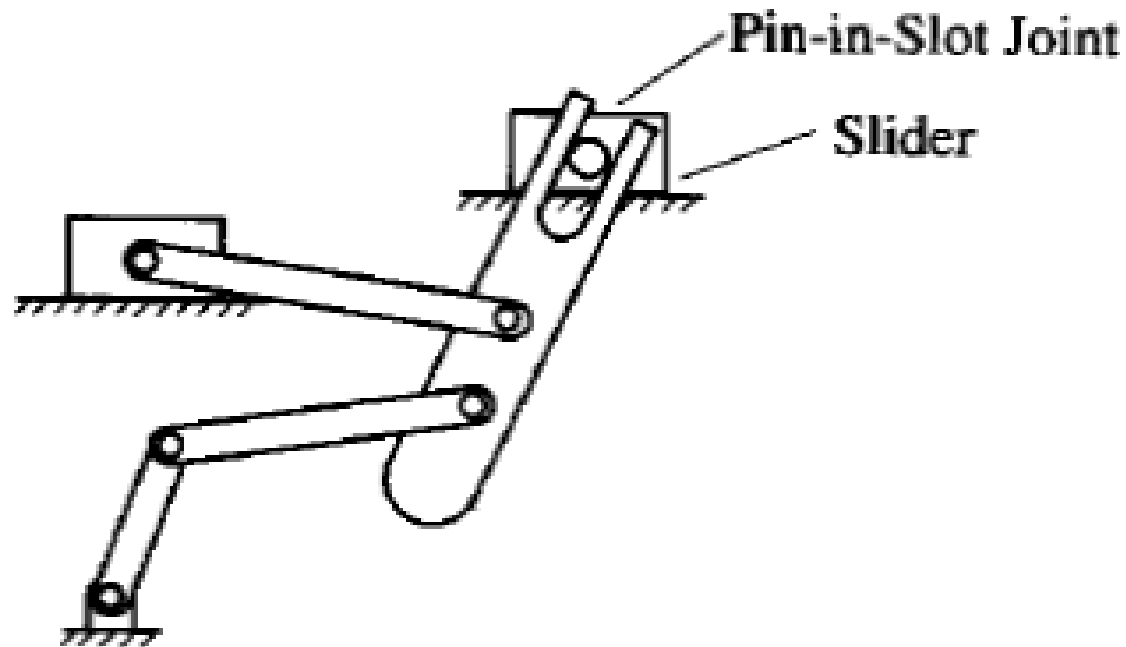
A pin in a slot is a J_2 joint. It has two degrees of freedom because it allows **rotation and sliding** between the two links (the pink link and the gray link in this figure.)

Example: Mobility



Use the Gruebler-Kutzbach criterion to calculate the mobility of this linkage.

Example: Mobility

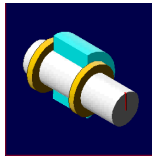


Use the Gruebler-Kutzbach criterion to calculate the mobility of this linkage.

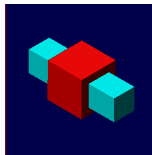
Common Examples of J_1 and J_2 joints

J_1 joints:

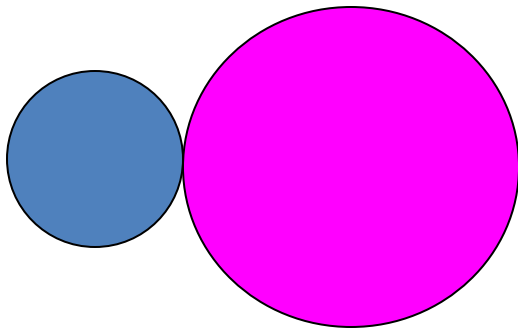
- Revolute (pin) joints



- Prismatic (slider) joints

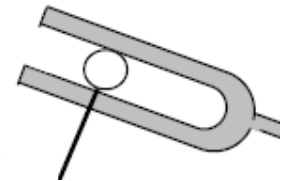


- Pure rolling contact

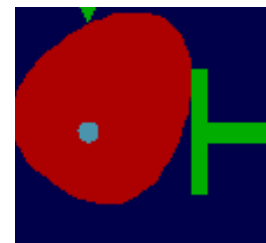


J_2 joints:

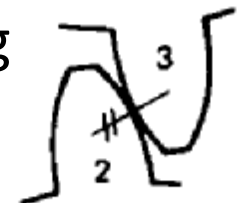
- Pin-in-slot



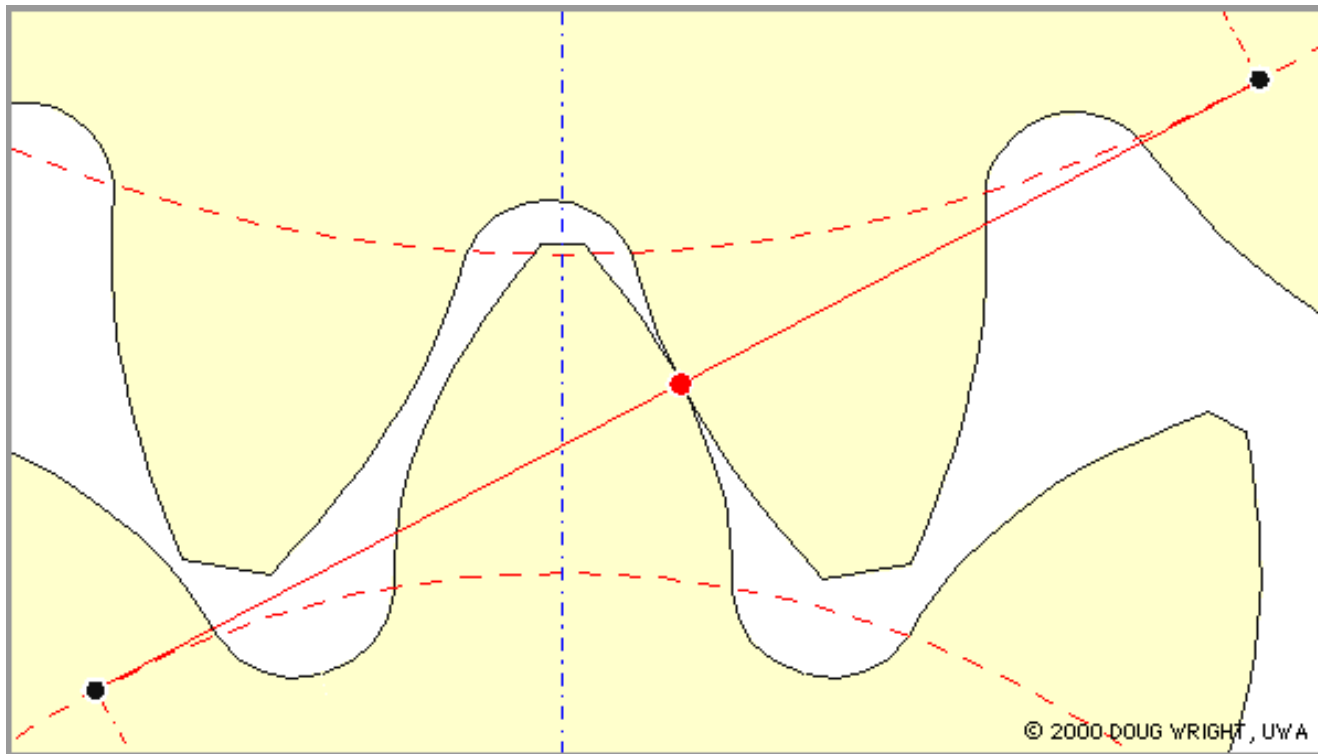
- Joints that allow rotation and sliding between the two links (includes some cam joints).



- 2 gears meshing

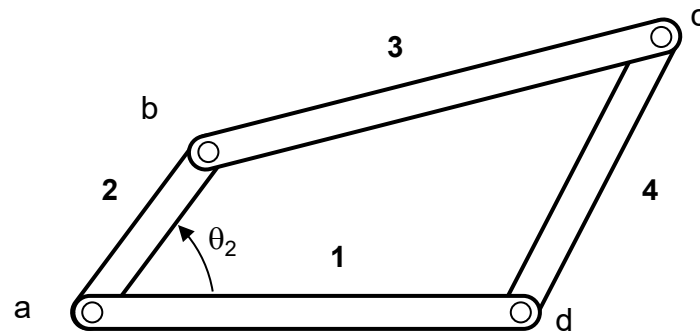
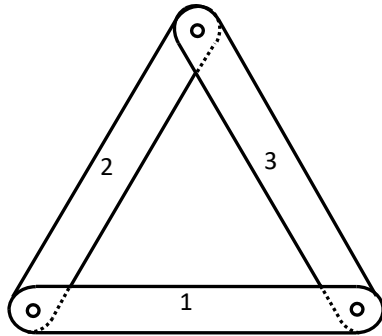


Gears: Rolling and Sliding contact between gear teeth



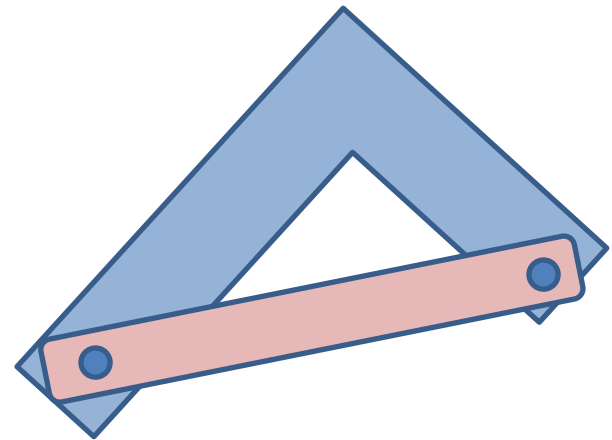
Intuitive Examples

$$M = 3(L - 1) - 2J_1 - J_2$$



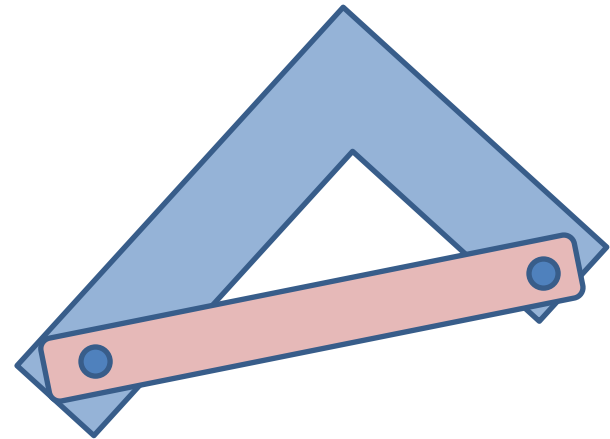
Preloaded Structures ($M < 0$)

$$M = 3(L-1) - 2J_1 - J_2$$



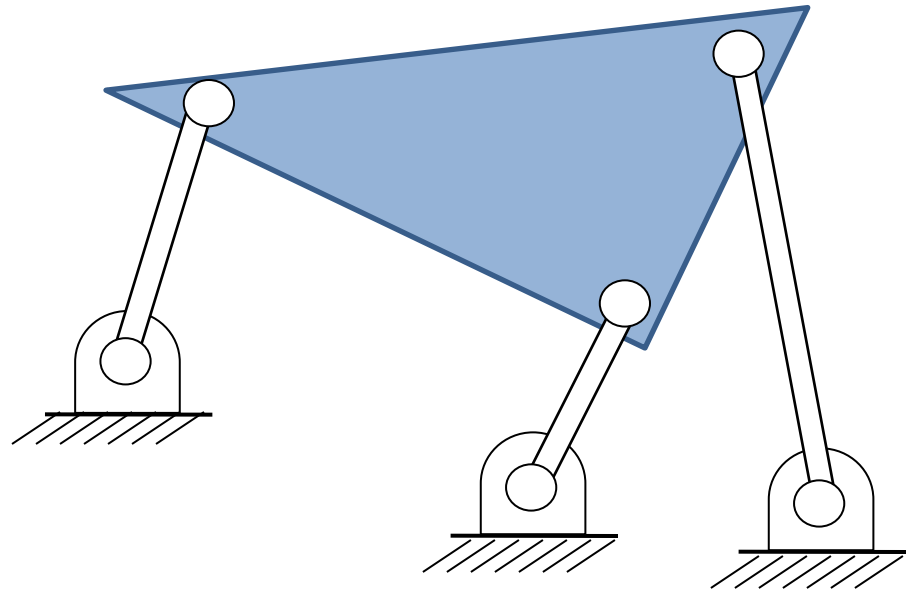
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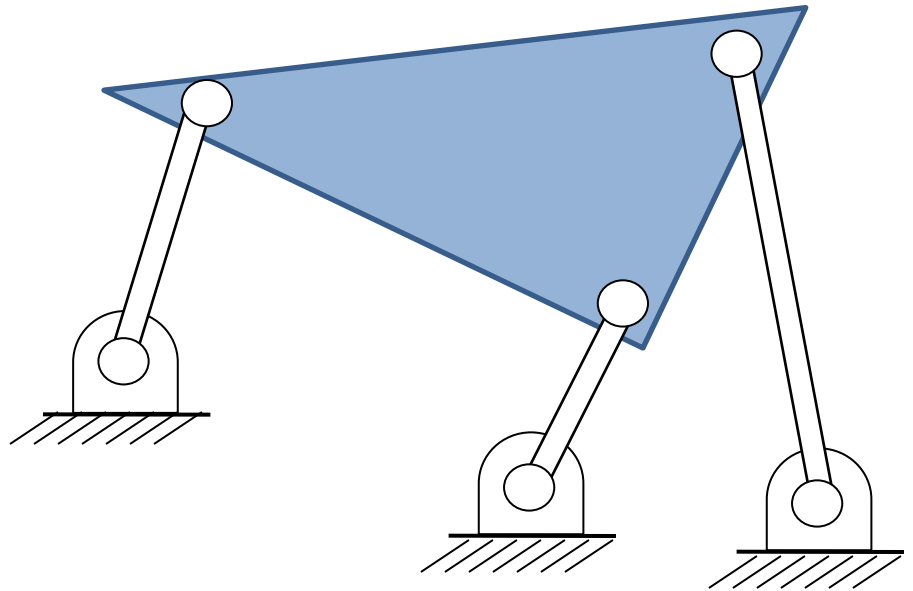
Calculate the mobility

$$M = 3(L - 1) - 2J_1 - J_2$$

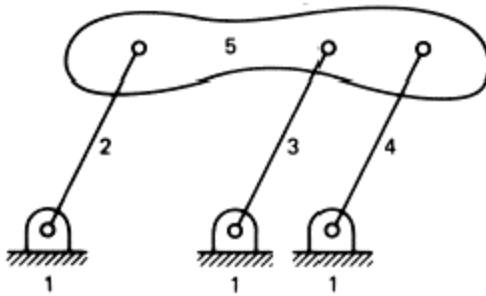


Calculate the mobility

$$M = 3(L - 1) - 2J_1 - J_2$$



Special Cases where the Gruebler-Kutzbach criterion gives incorrect results:



$$L = 5$$

$$J_1 = 6$$

$$J_2 = 0$$

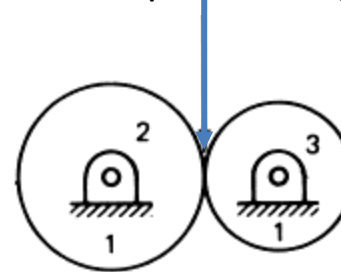
$$M = 3(L-1) - 2J_1 - J_2$$

$$= 3(5-1) - 2(6) - 0$$

$$= 0 \text{ according to Gruebler.}$$

BUT it can actually move due to the special geometrical configuration. (Links 2, 3, and 4 are parallel and exactly the same length.)

2 rollers with pure rolling contact



$$L = 3$$

$$J_1 = 3 \text{ (2 pin joints, \& pure rolling)}$$

$$J_2 = 0$$

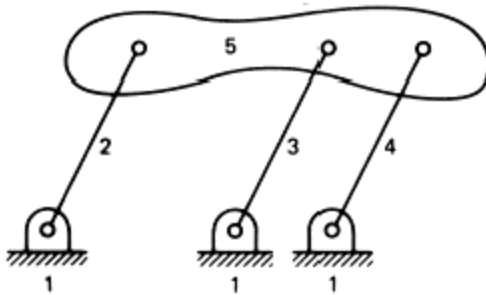
$$M = 3(L-1) - 2J_1 - J_2$$

$$= 3(3-1) - 2(3) - 0$$

$$= 0 \text{ according to Gruebler.}$$

BUT it can actually move due to the special geometrical configuration. (Exact spacing of rollers and ground pivots.)

Special Cases where the Gruebler-Kutzbach criterion gives incorrect results:



$$L = 5$$

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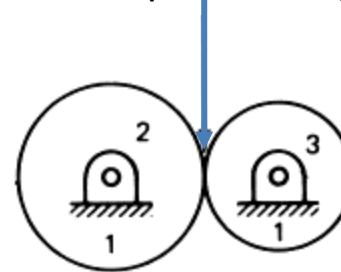
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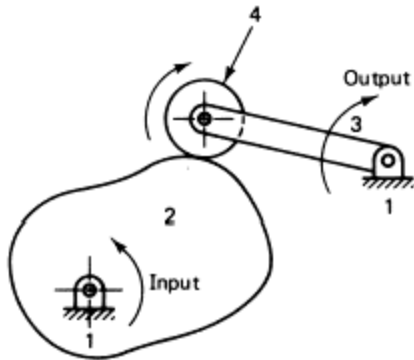
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BUT it can actually move due to the special geometrical configuration. (Exact spacing of rollers and ground pivots.)

Cam and Follower

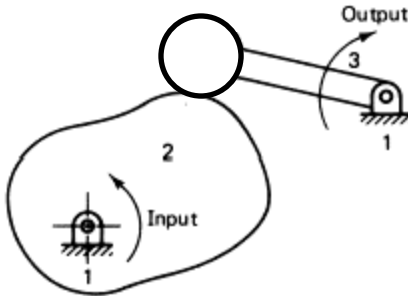
On any cam and follower problems, state your assumptions!



IF the cam joint allows rotation & sliding between links 2 and 4:

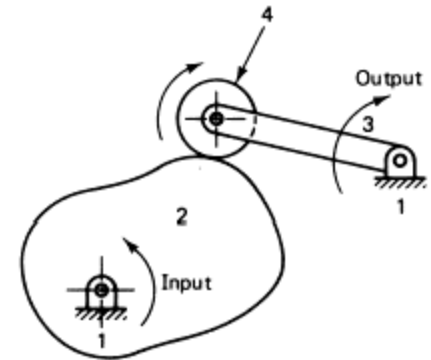
$$\begin{aligned} L &= 4 \\ J_1 &= 3 \\ J_2 &= 1 \\ M &= 3(L-1) - 2J_1 - J_2 \\ &= 3(4-1) - 2(3) - 1 \\ &= 2 \end{aligned}$$

(There is one redundant degree of freedom – slipping of the follower #4 along the surface of the cam.)



IF the follower is welded to the output, the joint still allows rotation and sliding between links 2 and 3 (which is now the link that contacts link 2.)

$$\begin{aligned} L &= 3 \\ J_1 &= 2 \\ J_2 &= 1 \text{ (See above)} \\ M &= 3(L-1) - 2J_1 - J_2 \\ &= 3(3-1) - 2(2) - 1 \\ &= 1 \end{aligned}$$

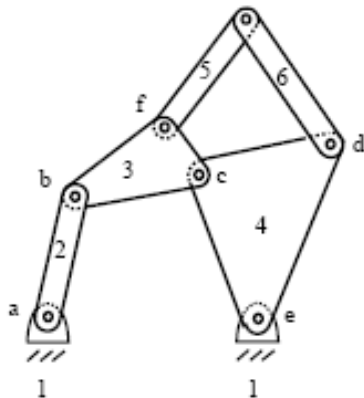
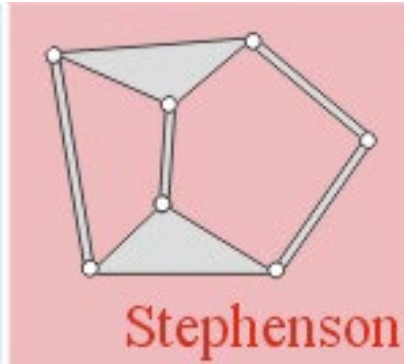
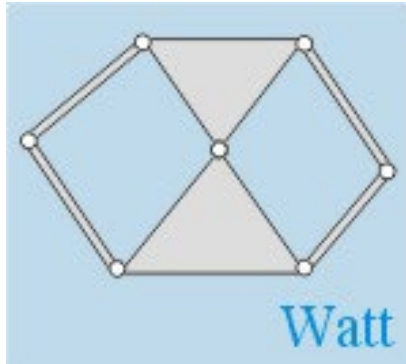


IF the follower is a roller and there is only pure rolling contact between links 2 and 4 (no slipping):

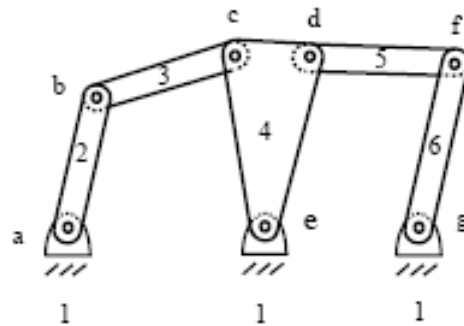
$$\begin{aligned} L &= 4 \\ J_1 &= 4 \\ J_2 &= 0 \\ M &= 3(L-1) - 2J_1 - J_2 \\ &= 3(4-1) - 2(4) - 0 \\ &= 1 \end{aligned}$$

(Same answer as the previous case.)

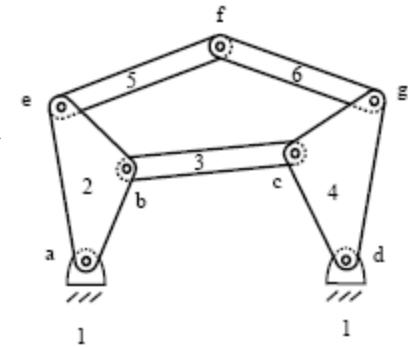
Six-bar linkages with ONE Degree of Freedom



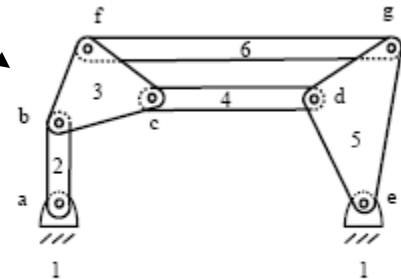
Watt 1



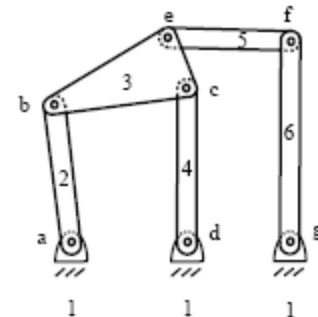
Watt 2



Stephenson 1



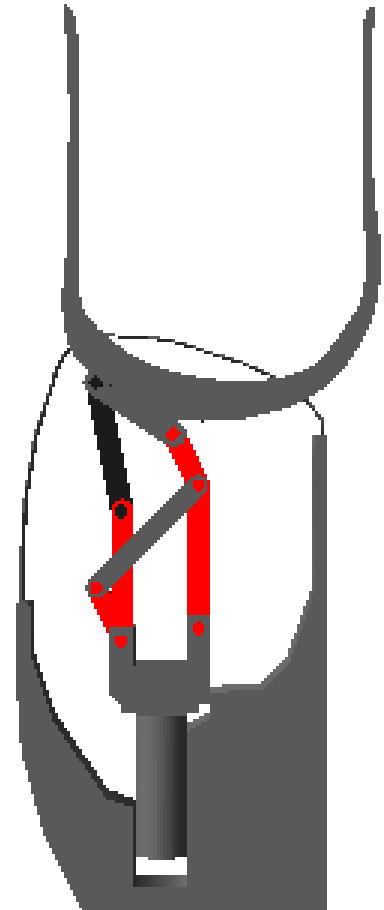
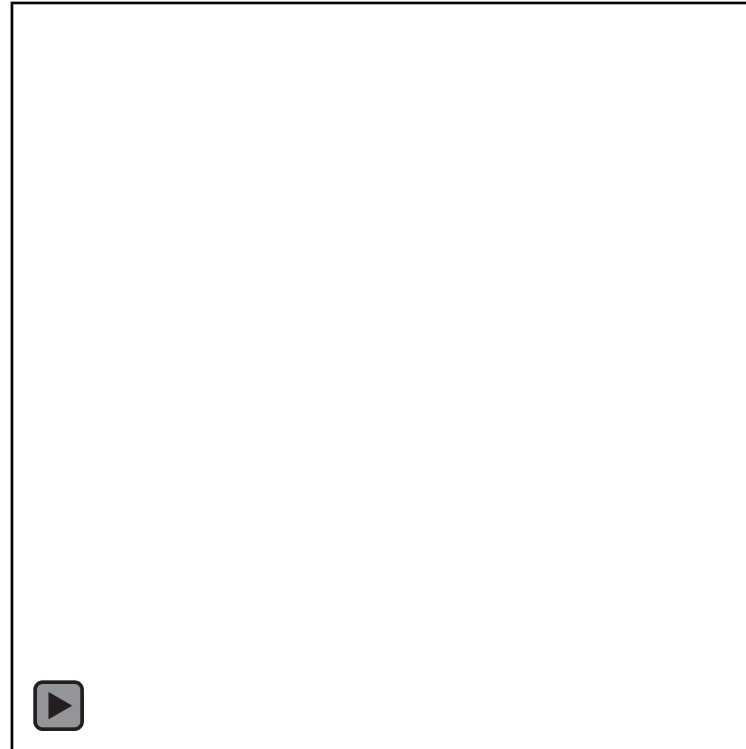
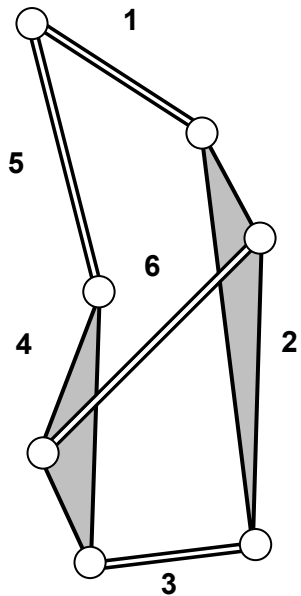
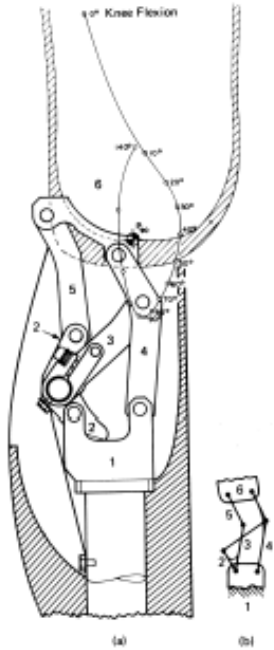
Stephenson 2



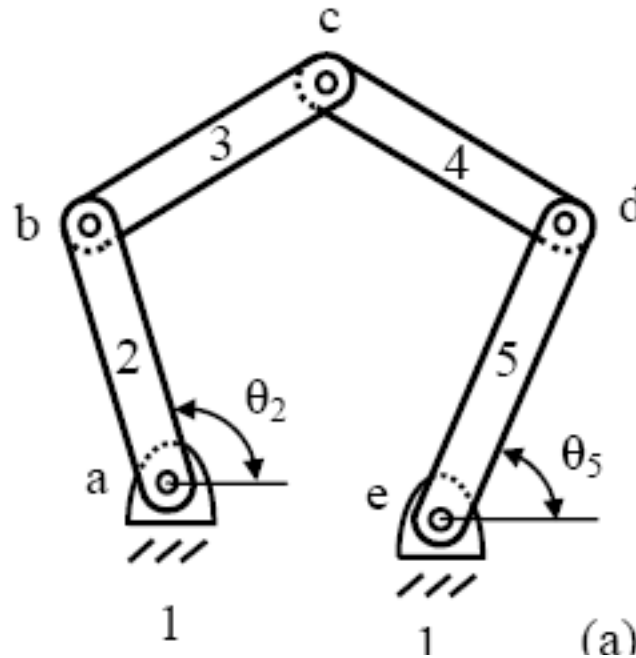
Stephenson 3

FIVE UNIQUE COMBINATIONS OF 2 ternary and 4 binary links

6-Bar Example: Prosthetic Knee



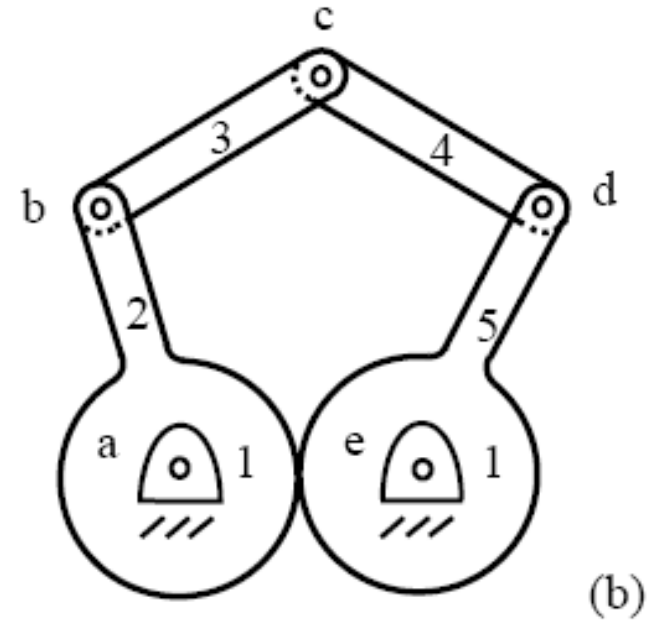
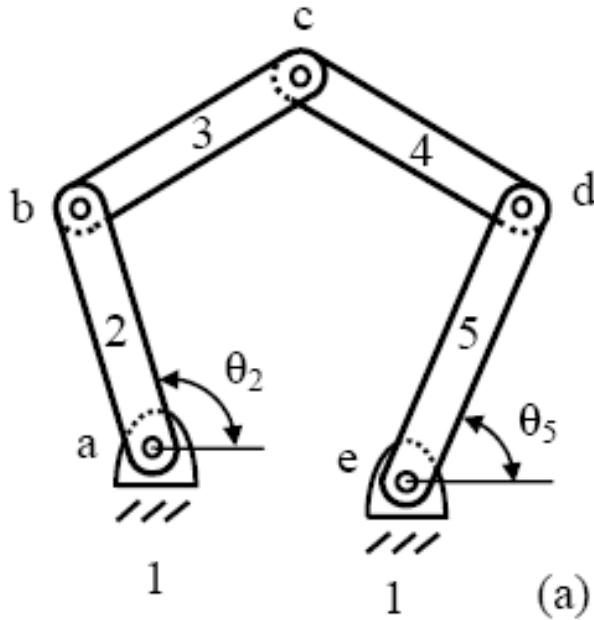
Why are five-bar linkages not commonly used?



$M = ?$

$$M = 3(L - 1) - 2J_1 - J_2 = 2$$

Geared Five-Bar Mechanism



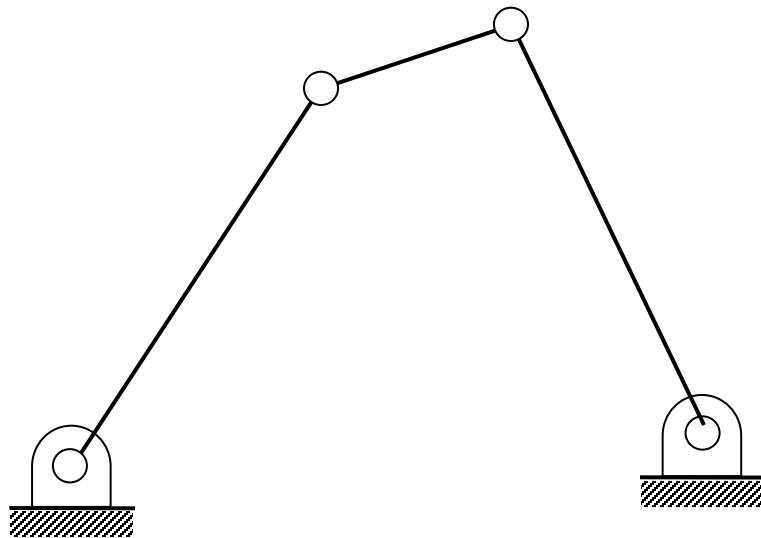
$$M = 3(L-1) - 2J_1 - J_2$$

$$= 1$$



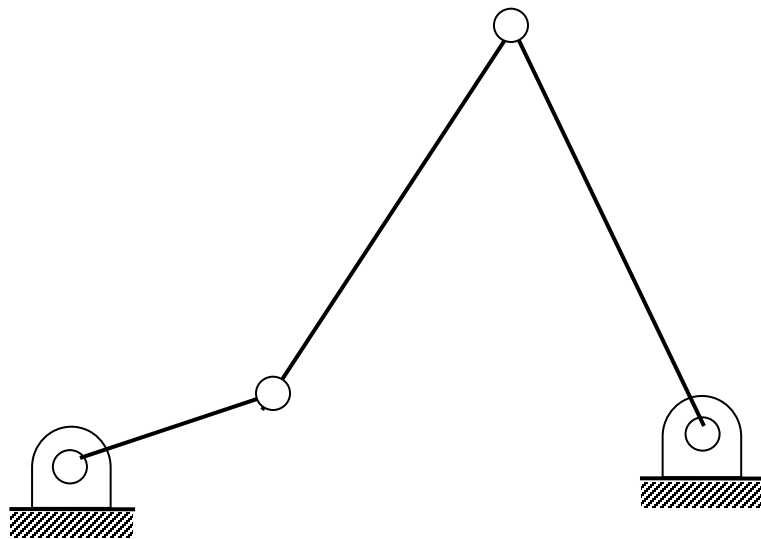
ME350 PUZZLE

Will any of the links in this mechanism be able to make a complete rotation?



ME350 PUZZLE

Will any of the links in this mechanism be able to make a complete rotation?





Stretch

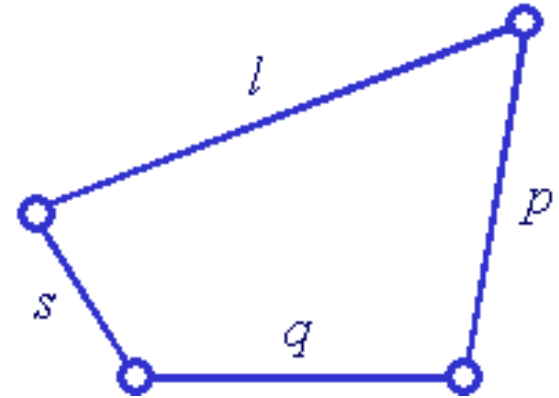
Grashof's Law (Grashof's Criterion)

Given: a four-bar linkage with

s = length of the shortest link

l = length of the longest link

p and q = lengths of the other two links

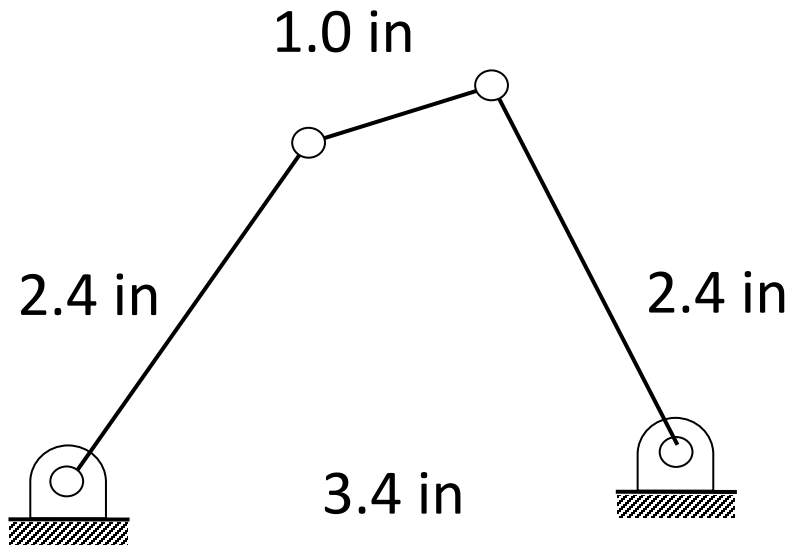


If $s + l < p + q$, then at least one link can make a complete 360 degree rotation relative to the ground plane.

This is called a **Grashof linkage**.

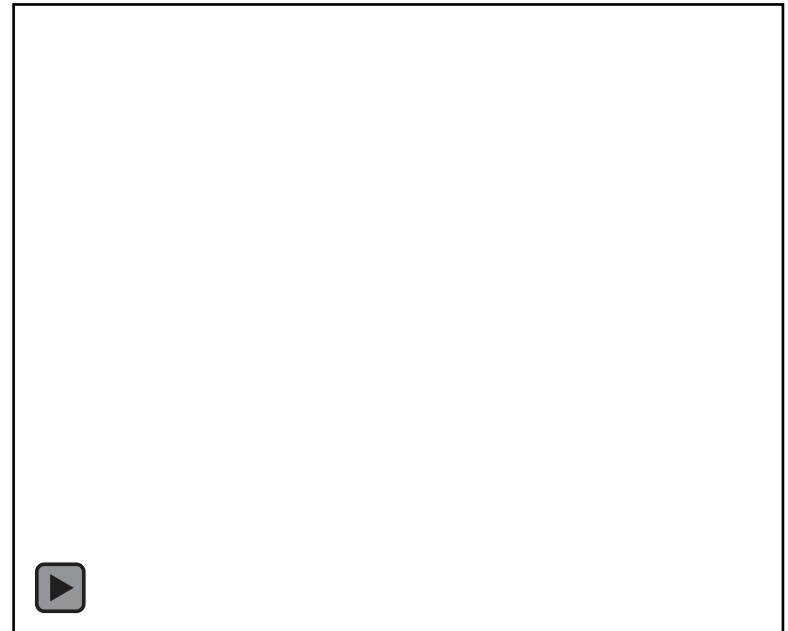
Example

Find the **Grashof condition (Grashof, Non-Grashof, or Change Point)** of this linkage. Link lengths are finally given.



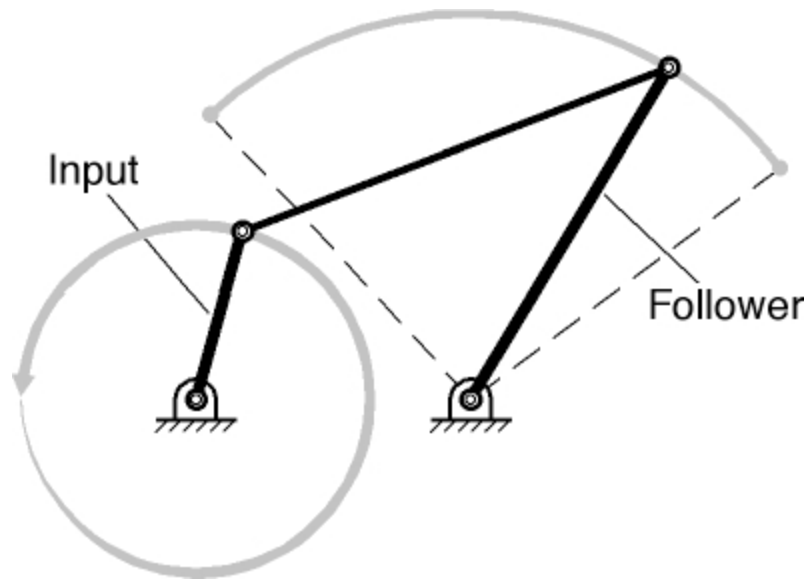
What is $S+L$? 4.4

What is $P+Q$? 4.8



Crank-Rocker Linkage

In this type of Grashof linkage, the input crank is adjacent to the ground link. The input crank rotates completely around, and the follower oscillates or “rocks” back and forth.



Kinematic Inversions of a Four-Bar Linkage

A kinematic inversion is created by grounding a different link in the linkage.

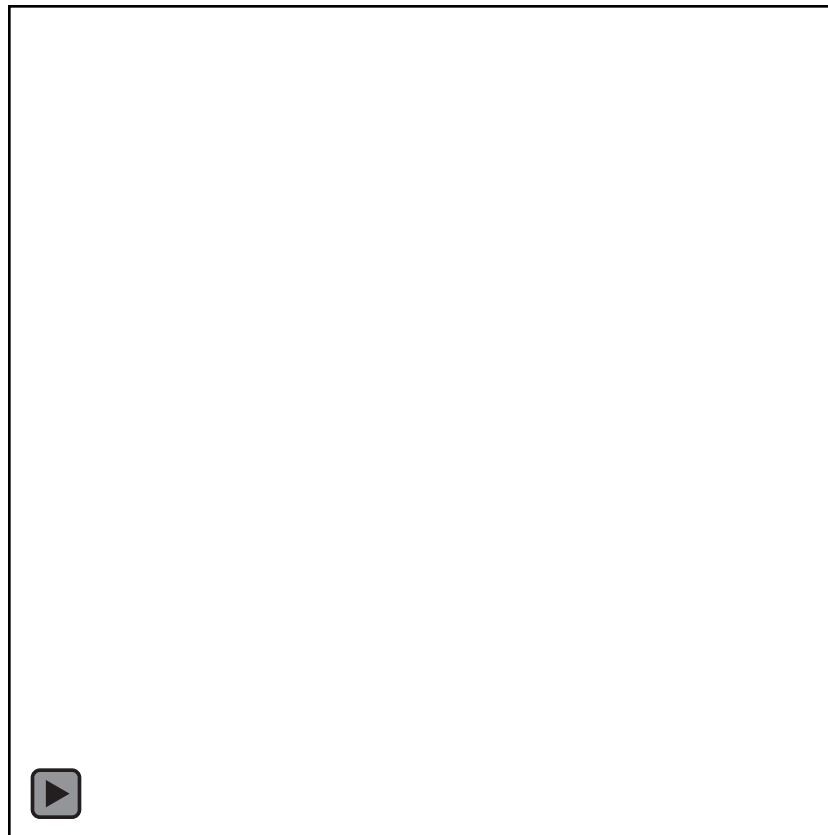
The relative motion between the links is the same in all inversions, while the absolute motion is different.

Crank-Rocker

Drag Link

Crank-Rocker
#2

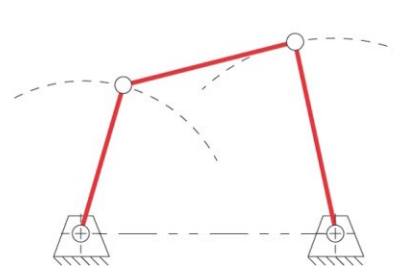
Double Rocker



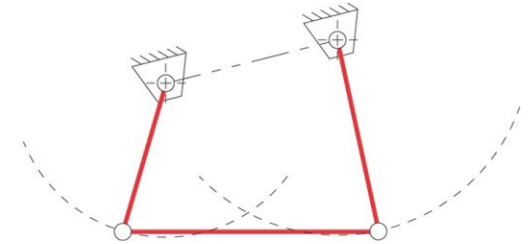
Non-Grashof Triple-Rocker Linkage

If $s + l > p + q$, it is a non-Grashof Triple Rocker linkage.

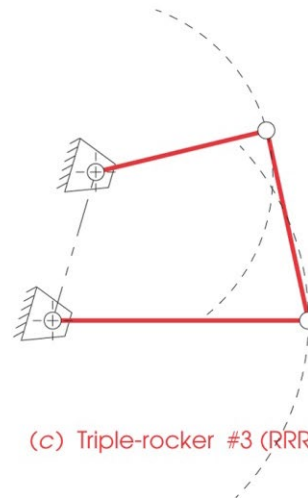
None of the links can rotate a full 360 degrees.



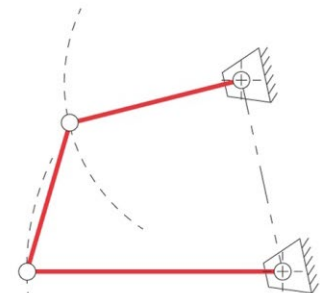
(a) Triple-rocker #1 (RRR1)



(b) Triple-rocker #2 (RRR2)



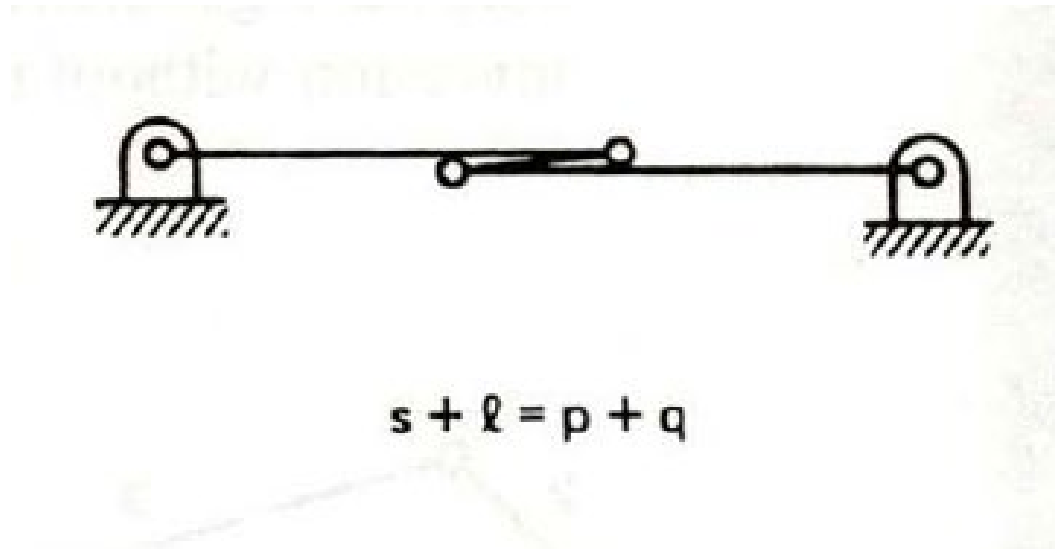
(c) Triple-rocker #3 (RRR3)



(d) Triple-rocker #4 (RRR4)

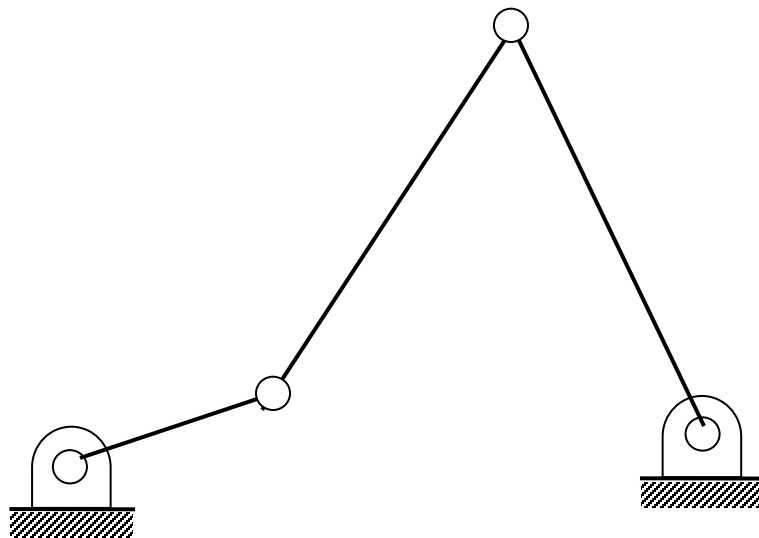
Change-Point Condition

If $s + l = p + q$, the linkage is similar to a Grashof linkage but suffers from the change-point condition (uncertainty or folding positions.)



Example #2

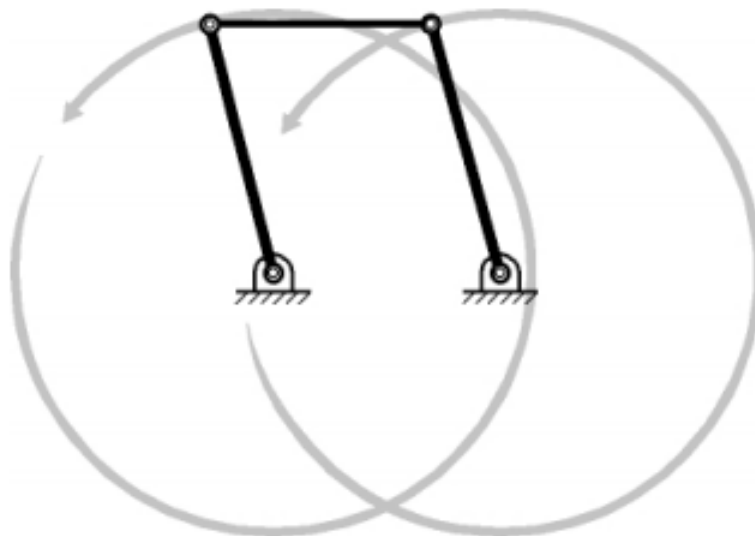
Find the **Grashof condition (Grashof, Non-Grashof, or Change Point)** of this linkage. Link lengths are the same as in the previous example, but two links are swapped.



Parallelogram Linkage

- lengths of the input and output links are equal
- lengths of the coupler and ground are also equal

Result: The input and output links rotate completely around, while the coupler remains parallel to the ground link.



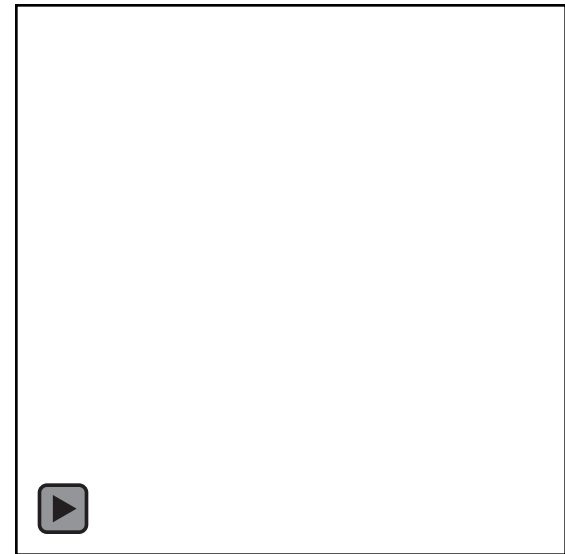
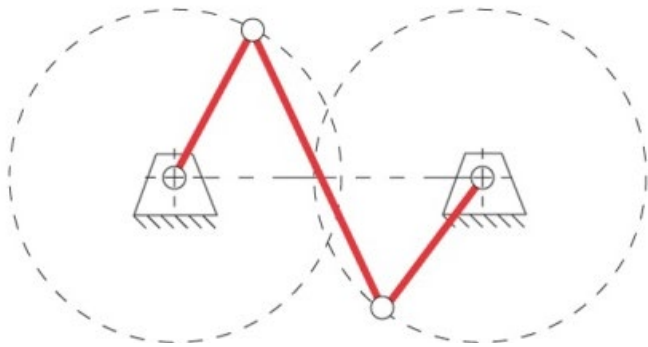
Application of a Parallelogram Linkage



<http://www.youtube.com/watch?v=jdPNOzN82so>

Parallelogram Linkage

The parallelogram linkage can also reach a position of crossed configuration called an anti-parallelogram linkage. It is an example of a change-point linkage.



The Moby Dick ride is an example of a change point linkage.

How could the designers of this ride change the design, to prevent the change point condition from occurring?

